1. **Time Saturn takes to complete one revolution around the Sun-**
   (a) 18.5 years (b) 36 years
   (c) 29.5 years (d) 84 years
   44th B.P.S.C. (Pre) 2000

Ans. (c)

Saturn takes 29.456 years to complete one revolution around the Sun. Hence, option (c) is the correct answer.

2. **Which of the following planets has rings around it –**
   (a) Saturn (b) Mars
   (c) Mercury (d) Earth
   U.P.P.C.S. (Pre) 1990

Ans. (a)

Saturn is especially famous for its peculiar feature of rings. A Total of seven rings of Saturn have been discovered. Its A,B, D and E rings can be seen from Earth. The colour of Saturn from Earth seems yellow. Besides Saturn, Jupiter, Uranus and Neptune also have rings.

3. **After seven years of its journey spacecraft Cassini started its revolution in June-2004 around which planet–**
   (a) Mercury (b) Jupiter
   (c) Mars (d) Saturn
   U.P.P.C.S. (Mains) 2004

Ans. (d)

On October 15, 1997 the Cassini spacecraft was launched to Saturn and began its revolution around this planet in June 2004.

4. **Planet Saturn –**
   (a) Is colder than Pluto (b) Is colder than Neptune
   (c) Warmer than Neptune (d) Warmer than Jupiter
   U.P.P.C.S. (Pre) 2010

Ans. (c)

The Saturn is warmer than Neptune. According to the distance sequence of planets from the Sun (Mercury-Venus-Earth-Mars-Jupiter-Saturn-Uranus-Neptune), Saturn is closer to the Sun than Neptune. Therefore Saturn will be warmer in comparison to Neptune, but Venus is warmer than Mercury due to high concentration of CO_2 in its atmosphere.

5. **Titan is the largest moon of the planet –**
   (a) Mars (b) Venus
   (c) Jupiter (d) Saturn
   U.P.U.D.A./L.D.A. (Pre) 2006

Ans. (d)

Titan is the largest moon (natural satellite) of Saturn. The diameter of Titan is 5,150 km. Prominent satellite of Saturn is Atlas, Epimetheus, Tethys, Lapetus, Dione, Phoebe, Helene and Enceladus, etc. Rhea is, second largest natural satellite of Saturn, having the diameter of about 1,528 km.

x. The Uranus, Neptune and Pluto

*Uranus is the 7th planet from the Sun and the third largest planet in the solar system. Uranus was discovered by Sir William Herschel on 13th March 1781. The axial tilt or equatorial inclination of Uranus is 97.77°. Like Venus, Uranus rotates east to west. Uranus takes about 17 hours and 14 minutes to rotate on its axis, and about 84 Earth years to complete an orbit of the Sun (a Uranian year). *On Uranus, the Sun rises in the west and sets in the east. *The composition of Uranus atmosphere is different from its bulk, consisting mainly of molecular hydrogen, helium, methane and ammonia etc.. *Due to its distance from the Sun, Uranus is an **ice giant** and appears to be blue-green in colour due to presence of methane in its atmosphere. Uranus has 28 moons. Just like Saturn, Uranus also has planetary rings.

*Neptune was discovered by German astronomer Johann Gottfried Galle on the night of 23-24 September 1846. One day on Neptune takes about 16 hours and 6 minutes (the time it takes for Neptune to rotate), and Neptune makes a complete orbit around the sun in about 164.9 years. Neptune's axis of rotation is tilted at **28.3°** with respect to the plane of its orbit around the sun.

*Neptune is the **coldest planet** in the solar system with an average temperature of -200°C. Neptune does not have a solid surface. Its atmosphere is mostly made up of hydrogen, helium and methane. *Neptune has 16 known natural satellites among these **Triton** and **Nereid** are the prominent one. *In 2006 when Pluto was excluded from the category of planets, Neptune became the farthest planet in the solar system.

*Pluto was discovered by Clyde Tombaugh in 1930. It was considered to be the smallest and the 9th planet of the solar system. *Between 14 to 25 August 2006 , the 26th General Assembly of the International Astronomical Union was held in **Prague** (Czech Republic). The assembly excluded Pluto from the list of planets and reclassified it as a dwarf planet. This was because the orbit of Pluto was overlapping with that of Neptune. *Pluto has five known natural satellites. The closest to Pluto is **Charon**, followed by **Styx**, **Nix**, **Kerberos** and **Hydra**. Charon is the largest natural 'satellite' of Pluto. Pluto takes about 247.9 years to complete one revolution around the sun.

1. **Which of the following statements are correct regarding the solar system?**
   1. **Mercury is the hottest planet in the solar system.**
   2. **Ganymede, satellite of Saturn, is the largest satellite in the solar system.**
   3. **Neptune is surrounded by methane gas rings of sub-zero temperature.**
   4. **Phobos and Deimos are two satellites of Mars**

   (a) Only 1 and 2 are correct
   (b) Only 2 and 3 are correct

(c) Only 3 and 4 are correct

(d) 1, 2, 3 and 4 all are correct

Chhattisgarh P.C.S. (Pre) 2020

Ans. (c)

The hottest planet in the Solar System is Venus. Its mean temperature is 464^{0}C. Jupiter's satellite Ganymede is the largest Satellite in the Solar System. Phobos and Deimos are the two satellites of Mars. Neptune is surrounded by rings of methane gas at subzero temperatures. Due to this, the colour of Neptune appears to be blue-green.

2. For one revolution around the Sun, Uranus takes –

(a) 84 years (b) 36 years

(c) 18 years (d) 48 years

44th B.P.S.C. (Pre) 2000

Ans. (a)

Uranus's period of revolution around the Sun is 84 years. A day on Uranus is approximately 17 hours and 14 minutes.

3. The year is largest on –

(a) Pluto (b) Jupiter

(c) Neptune (d) Earth

39th B.P.S.C. (Pre) 1994

Ans. (a)

The year is largest on Pluto because it is farthest among the other celestial objects from the Sun. Pluto takes a period of 247.9 Earth years to complete one revolution around the Sun.

4. Which planet takes the longest period in revolving around the Sun ?

(a) Uranus (b) Jupiter

(c) Neptune (d) Pluto

U.P. Lower Sub. (Spl) (Pre) 2004

Ans. (d)

Neptune takes the longest period to make a revolution around the Sun. As we know that planet Pluto has been removed from the category of planets since 2006. Now starting from Mercury up to Neptune only 8 planets come into the category of true planets. The time period of revolution of all four planets is given below –

(Celestial Body)	**(Time of Revolution in years)**
Jupiter	11.86 (12)
Uranus	83.7 (84)
Neptune	164.9 (165)
Pluto	247.9 (248)

5. According to a new definition adopted by 'International Astronomical Union' in 2006, which of the following is not a 'planet' ?

(a) Uranus (b) Neptune

(c) Pluto (d) Jupiter

M.P.P.C.S. (Pre) 2012

Ans. (c)

In the 26th General Assembly of the International Astronomical Union held from 14-25 August, 2006, Pluto was removed from the category of planets and was assigned the status of a dwarf planet.

6. Which is the smallest planet of the solar system ?

(a) Pluto (b) Mars

(c) Venus (d) Mercury

U.P.P.C.S. (Pre) 1991

Ans. (a)

According to the previous assumption, Pluto was considered as the smallest planet with a diameter of 2376 km. But now we can consider planet Mercury as the smallest planet of the solar system. The diameters of all eight planets are given below :

Mercury	-	4,879 km
Venus	-	12,104 km
Earth	-	12,756 km
Mars	-	6,792 km
Jupiter	-	142,984 km
Saturn	-	120,536 km
Uranus	-	51,118 km
Neptune	-	49,528 km

7. The coldest planet in the solar system is –

(a) Neptune (b) Jupiter

(c) Mars (d) Saturn

M.P.P.C.S. (Pre) 2014

Ans. (a)

Neptune which is farthest from the Sun, is the coldest planet of our solar system. The temperature of Neptune's surface is –200º C and its atmosphere is made up of hydrogen, helium and methane.

8. Which of the following planets is the farthest planet of the solar system –

(a) Neptune

(b) Pluto

(c) Some times Neptune and some times Pluto

(d) Mars

I.A.S. (Pre) 2002

I.A.S. (Pre) 2005

Ans. (c)